

Systematic Trading Strategies with Machine Learning Algorithms

Latent Variable Models in Financial Asset Regime Detection



April 29, 2026

From GMMs to HMMs

- Recap: Gaussian Mixture Models

- Introducing Hidden Markov Models

- An example: Discrete HMMs

Estimation Problems

- Objectives

- The Filtering-Smoothing Probabilities

- The Learning Problem

- Predicting the next hidden state

- Forecasting Market Turbulence Regimes

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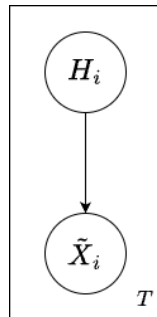
The Filtering-Smoothing Probabilities

The Learning Problem

Predicting the next hidden state

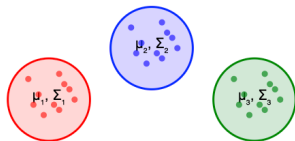
Forecasting Market Turbulence Regimes

- ▶ Goal: Represent complex, multimodal distributions using a mixture of Gaussians
- ▶ Latent variable:
 $H_i \sim \mathcal{M}(1, \pi_1, \dots, \pi_M)$
- ▶ Observation:
 $\tilde{X}_i | H_i = m \sim \mathcal{N}(\mu_m, \Sigma_m)$
- ▶ EM algorithm used to estimate parameters $\theta = (\pi, \mu, \Sigma)$

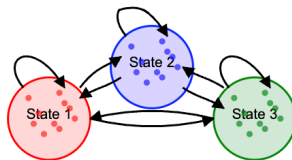


- ▶ GMMs assume hidden variables $H_1, \dots, H_T$ are i.i.d.
- ▶ Ignores temporal structure in sequential data
- ▶ Need a model where:
 - ▶ Hidden states evolve over time
 - ▶ Observations depend on the current hidden state
- ▶ This leads to **Hidden Markov Models (HMMs)**

Gaussian Mixture Model (GMM)



Hidden Markov Model (HMM)



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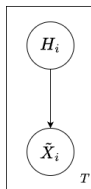
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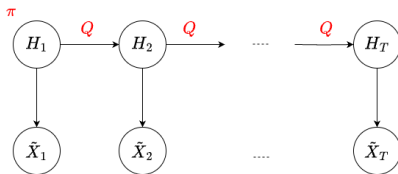
Predicting the next hidden state

Forecasting Market Turbulence Regimes

- ▶ Latent states: $H_1, \dots, H_T$ with M possible hidden states.
- ▶ Observations: $\tilde{X}_1, \dots, \tilde{X}_T \in \mathbb{R}^d$ or $\mathcal{O} = \{o_1, \dots, o_D\}$
- ▶ **Markov property (transition model):**
 - ▶ The hidden state sequence forms a first-order Markov chain.
- ▶ **Emission independence:**
 - ▶ Observations are conditionally independent given the current hidden state:

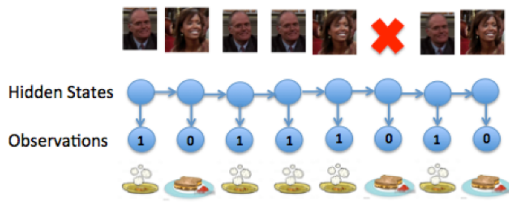


Gaussian
Mixture Model



Hidden Markov Model

- ▶ Observation space: $\mathcal{O} = \{o_1, \dots, o_D\}$
- ▶ **Hidden state dynamics:**
 - ▶ Initial distribution: $\pi_m = P(H_1 = m)$
 - ▶ Transition matrix: $Q_{ij} = P(H_{t+1} = j \mid H_t = i)$
- ▶ **Emission distribution:**
 - ▶ Emission matrix: $O_{mj} = P(\tilde{X}_t = o_j \mid H_t = m)$
- ▶ Parameter set:
$$\theta = (\pi, Q, O)$$
- ▶ Example from Programming Session 3:



- ▶ Number of hidden states: M

- ▶ **Hidden state dynamics:**

- ▶ Initial distribution:

$$\pi_m = P(H_1 = m)$$

- ▶ Transition matrix:

$$Q_{ij} = P(H_{t+1} = j \mid H_t = i)$$

- ▶ **Emission model (continuous):**

$$\tilde{X}_t \mid H_t = m \sim \mathcal{N}_d(\mu_m, \Sigma_m)$$

- ▶ Parameter set:

$$\theta = (\pi, Q, \mu, \Sigma)$$

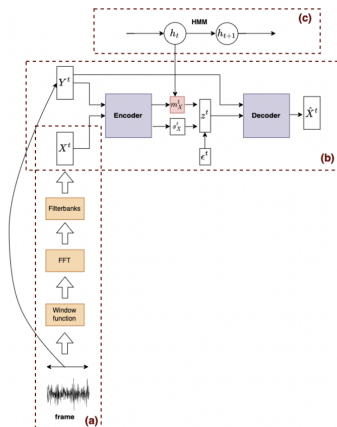


Figure: Forecasting Market Turbulence Regimes

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Who Ate Ross's Sandwich?

Dr. Ross Geller, a paleontologist at New York University, faces a peculiar dilemma.

- ▶ Every day, he brings a sandwich and stores it in the department refrigerator.
- ▶ His sandwich frequently disappears before lunch.
- ▶ He records what he can observe:
 - ▶ **0**: Sandwich is safe
 - ▶ **1**: Sandwich is missing



Hidden States (Unobserved):

- ▶ **State 0:** Dr. Donald is present — 90% chance sandwich is eaten
- ▶ **State 1:** Dr. Charlie is present — 50% chance sandwich is eaten
- ▶ **State 2:** Neither is present — 0% chance sandwich is eaten

State 0



State 1



State 2



Observation space:

$\mathcal{O} = \{0 : \text{Safe}, 1 : \text{Missing}\}$

Emission matrix O :

$$O = \begin{bmatrix} 0.1 & 0.9 \\ 0.5 & 0.5 \\ 1.0 & 0.0 \end{bmatrix}$$



$$\begin{pmatrix} 0.1 \\ 0.9 \end{pmatrix}$$

$$\begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$$

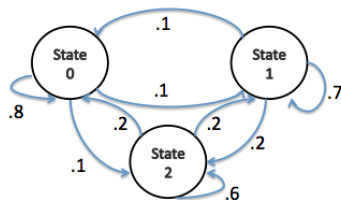
$$\begin{pmatrix} 1. \\ 0. \end{pmatrix}$$

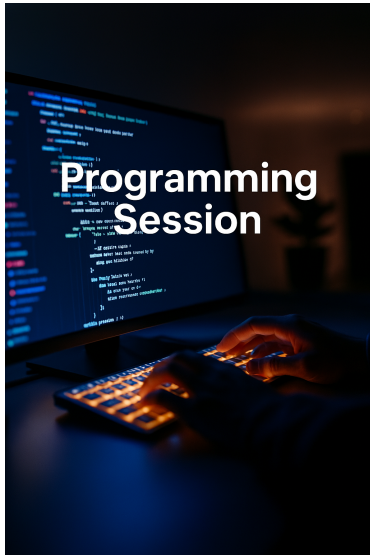
Initial state distribution (uniform):

$$\pi = \left[\frac{1}{3}, \frac{1}{3}, \frac{1}{3} \right]$$

Transition matrix Q :

$$Q = \begin{bmatrix} 0.8 & 0.1 & 0.1 \\ 0.1 & 0.7 & 0.2 \\ 0.2 & 0.2 & 0.6 \end{bmatrix}$$





Programming Session 3: Section 1

- ▶ Section 1: Create the Synthetic Data
- ▶ *Click here to access the programming session*

Solution will be posted tonight on the GitHub page.

- ▶ *Click here to access the GitHub Page*

Feedback Poll

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Goal: Compute the likelihood of the observed sequence $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_T)$ given the model parameters θ .

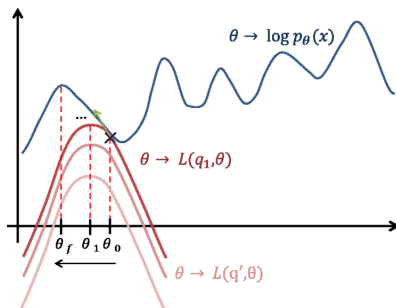
We illustrate this in the case of **continuous emissions**, where each hidden state emits a multivariate Gaussian.

$$p_{\theta}(\tilde{\mathbf{x}}) = \sum_{h_1} \sum_{h_2} \cdots \sum_{h_T} \pi_{h_1} \prod_{t=1}^{T-1} Q_{h_t, h_{t+1}} \prod_{t=1}^T \mathcal{N}(\tilde{x}_t; \mu_{h_t}, \Sigma_{h_t})$$

This computation is intractable due to the exponential number of hidden state paths (M^T sequences).

Approach: Use a recursive and efficient approach (**The Forward Algorithm**).

Approach: Expectation-Maximization (EM) algorithm:



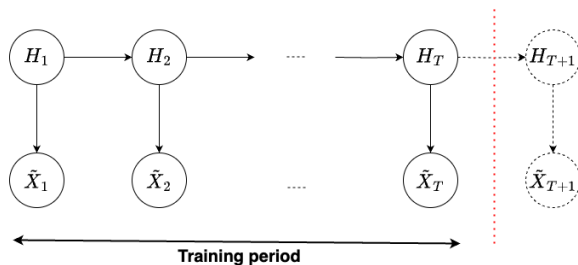
Objective 3: Predicting the Next Hidden State



Goal: Compute the distribution over hidden states at time $T + 1$ given past observations and learned parameters.

Approach: Once the model is trained using the EM algorithm, we can use the filtering probabilities $[\xi(T, h')]_{1 \leq h' \leq M}$ (which will be introduced later) to make the prediction:

$$p(H_{T+1} = h \mid \tilde{X}_1 = \tilde{x}_1, \dots, \tilde{X}_T = \tilde{x}_T) \quad \forall h \in \llbracket 1, M \rrbracket$$



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To solve these problems, we introduce the following probabilities:

- ▶ **Filtering probabilities:** $\xi \in \mathbb{R}^{T \times M}$

$$\begin{aligned}\forall (t, h) &\in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket, \\ \xi(t, h) &:= p(H_t = h \mid \tilde{X}_1, \dots, \tilde{X}_t)\end{aligned}$$

- ▶ **Smoothing probabilities:**

- ▶ $\psi \in \mathbb{R}^{T \times M}$

$$\begin{aligned}\forall (t, h) &\in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket, \\ \psi(t, h) &:= p(H_t = h \mid \tilde{X}_1, \dots, \tilde{X}_T)\end{aligned}$$

- ▶ $\phi \in \mathbb{R}^{T-1 \times M \times M}$

$$\begin{aligned}\forall (t, h, h') &\in \llbracket 1, T-1 \rrbracket \times \llbracket 1, M \rrbracket^2, \\ \phi(t, h, h') &:= p(H_t = h, H_{t+1} = h' \mid \tilde{X}_1, \dots, \tilde{X}_T)\end{aligned}$$

- **Goal:** Compute filtering probabilities:

$$\forall (t, h) \in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket, \quad \xi(t, h) := p(H_t = h \mid \tilde{X}_1, \dots, \tilde{X}_t)$$

- We introduce the **alpha variables**, which correspond to the joint probability of the observed sequence up to time t and the hidden state at time t :

$$\forall (t, h) \in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket, \quad \alpha(t, h) := p(\tilde{X}_1, \dots, \tilde{X}_t, H_t = h)$$

- We also introduce the **emission tensor** $\Gamma(t)$, a diagonal matrix encoding the likelihood of the current observation conditioned on each possible hidden state:

$$\forall t \in \llbracket 1, T \rrbracket : \Gamma(t) := \begin{pmatrix} p(\tilde{X}_t \mid H_t = 1) & & \\ & \ddots & \\ & & p(\tilde{X}_t \mid H_t = M) \end{pmatrix} \in \mathbb{R}^{M \times M}$$

► **Vector notation:**

$$\alpha_t = (\alpha(t, 1), \dots, \alpha(t, M))^T \in \mathbb{R}^M$$

$$\xi_t = (\xi(t, 1), \dots, \xi(t, M))^T \in \mathbb{R}^M$$

► **Key results:** *Click here for the detailed calculations*

- Forward Propagation: Recursive calculation of α variables:

$$\alpha_1 = \Gamma(1)\pi, \quad \forall t \geq 2 : \alpha_t = \Gamma(t)Q^T \alpha_{t-1}$$

- Filtering probabilities from alpha:

$$\forall t \in \llbracket 1, T \rrbracket \quad \xi_t = \frac{\alpha_t}{\mathbb{1}_M^T \alpha_t}$$

- Solving Objective 1: $p(\tilde{\mathbf{x}}) = \mathbb{1}_M^T \alpha_T$

- **Goal:** Compute smoothing probabilities:

$$\forall (t, h) \in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket, \quad \psi(t, h) := p(H_t = h \mid \tilde{X}_{1:T})$$

$$\forall (t, h, h') \in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket^2, \quad \phi(t, h, h') := p(H_t = h, H_{t+1} = h' \mid \tilde{X}_{1:T})$$

- We introduce the **beta variables**, which represent the likelihood of future observations given the current state:

$$\forall (t, h) \in \llbracket 1, T \rrbracket \times \llbracket 1, M \rrbracket, \quad \beta(t, h) := p(\tilde{X}_{t+1:T} \mid H_t = h)$$

- Vector / Matrix notation:

$$\beta_t = (\beta(t, 1), \dots, \beta(t, M))^T \in \mathbb{R}^M$$

$$\psi_t = [\psi(t, h)]_h \in \mathbb{R}^M, \quad \phi_t = [\phi(t, h, h')]_{h, h'} \in \mathbb{R}^{M \times M}$$

- ▶ **Key results:** *Click here for the detailed calculations*

- ▶ Backward Propagation: Recursive calculation of β variables:

$$\beta_T = \mathbb{1}_M, \quad \forall t \leq T-1 : \beta_t = Q\Gamma(t+1)\beta_{t+1}$$

- ▶ Smoothing probability using α and β :

$$\forall t \in \llbracket 1, T \rrbracket \quad \psi_t = \frac{\alpha_t \circ \beta_t}{\mathbb{1}_M^T \alpha_T}$$

- ▶ Joint smoothing probability:

$$\forall t \in \llbracket 1, T \rrbracket \quad \phi_t = \frac{\text{diag}(\alpha_t)Q\Gamma(t+1)\text{diag}(\beta_{t+1})}{\mathbb{1}_M^T \alpha_T}$$

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Algorithm EM Algorithm

Require: Observations $\tilde{X} = \{\tilde{x}_1, \dots, \tilde{x}_T\}$

Ensure: Optimal parameters θ

- 1: **Initialization:** Choose initial parameters $\theta^{(0)}$.
- 2: **while** not converged **do**
- 3: **E-step:** Update q to maximize the lower bound with respect to q

$$q_{t+1} \in \arg \max_q (\mathcal{L}(q, \theta_t))$$

- 4: **M-step:** Update θ to maximize the lower bound with respect to θ

$$\theta_{t+1} \in \arg \max_{\theta} (\mathcal{L}(q_{t+1}, \theta))$$

- 5: **end while**
 - 6: **return** Optimized parameters θ^*
-

- ▶ We apply EM to estimate parameters of an HMM given observations $\tilde{X}_{1:T}$.
- ▶ The E-step computes the expected complete log-likelihood with respect to the posterior distribution over hidden states:

$$\mathbb{E}_{\mathbf{h}|\tilde{\mathbf{x}}}[\log p_{\theta}(\mathbf{h}, \tilde{\mathbf{x}})]$$

- ▶ At iteration i , we decompose the complete log-likelihood as:

$$\log(p_{\theta^{(i)}}(\mathbf{h}, \tilde{\mathbf{x}})) = \log(\pi_{h_1}^{(i)}) + \sum_{t=1}^{T-1} \log(Q_{h_t, h_{t+1}}^{(i)}) + \sum_{t=1}^T \log(\mathcal{N}(\tilde{X}_t; \mu_{h_t}^{(i)}, \Sigma_{h_t}^{(i)}))$$

The expected log-likelihood is computed with respect to the posterior distribution:

$$\mathbb{E}_{H|\tilde{\mathbf{x}}}[\log(\pi_{h_1})] = \sum_{h=1}^M \log(\pi_h) p(H_1 = h | \tilde{\mathbf{x}})$$

$$\mathbb{E}_{H|\tilde{\mathbf{x}}}[\log(Q_{h_t, h_{t+1}})] = \sum_{h=1}^M \sum_{h'=1}^M \log(Q_{hh'}) p(H_t = h, H_{t+1} = h' | \tilde{\mathbf{x}})$$

$$\mathbb{E}_{H|\tilde{\mathbf{x}}}[\log(\mathcal{N}(\tilde{X}_t; \mu_{h_t}, \Sigma_{h_t}))] = \sum_{h=1}^M \log(\mathcal{N}(\tilde{X}_t; \mu_h, \Sigma_h)) p(H_t = h | \tilde{\mathbf{x}})$$

- ▶ We use the smoothing probabilities ψ and ϕ , which are computed using the **Forward-Backward algorithm**.
- ▶ This allows us to compute the expected complete log-likelihood w.r.t the posterior distribution:

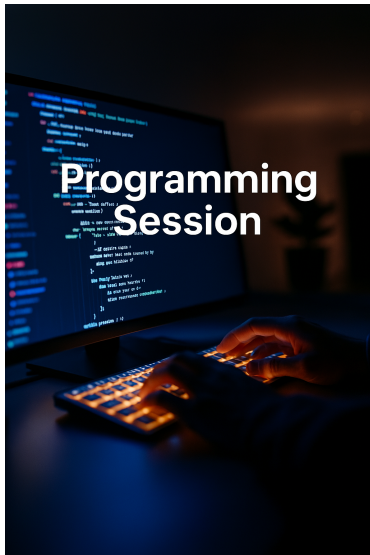
$$\begin{aligned}\mathbb{E}_{\mathbf{h}|\tilde{\mathbf{x}}}[\log p_{\theta}(\mathbf{h}, \tilde{\mathbf{x}})] &= \sum_{h=1}^M \log(\pi_h) \psi(1, h) + \sum_{t=1}^{T-1} \sum_{h, h'} \log(Q_{hh'}) \phi(t, h, h') \\ &\quad + \sum_{t=1}^T \sum_{h=1}^M \log(\mathcal{N}(\tilde{X}_t; \mu_h, \Sigma_h)) \psi(t, h)\end{aligned}$$

- ▶ *[Click here for the detailed calculations](#)*

- ▶ Maximize the expected log-likelihood w.r.t. parameters θ .
- ▶ Parameter update rules:

$$\begin{aligned}\pi_h^{(i+1)} &= \psi(1, h) \\ Q_{h,h'}^{(i+1)} &= \frac{\sum_{t=1}^{T-1} \phi(t, h, h')}{\sum_{t=1}^{T-1} \psi(t, h)} \\ \mu_h^{(i+1)} &= \frac{\sum_{t=1}^T \psi(t, h) \tilde{X}_t}{\sum_{t=1}^T \psi(t, h)} \\ \Sigma_h^{(i+1)} &= \frac{\sum_{t=1}^T \psi(t, h) (\tilde{X}_t - \mu_h^{(i)}) (\tilde{X}_t - \mu_h^{(i)})^T}{\sum_{t=1}^T \psi(t, h)}\end{aligned}$$

- ▶ *Click here for the detailed calculations*



Programming Session 3: Section 2

- ▶ Section 2: The Learning Problem - EM Algorithm
- ▶ *Click here to access the programming session*

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- ▶ *Click here to access the GitHub Page*

Feedback Poll

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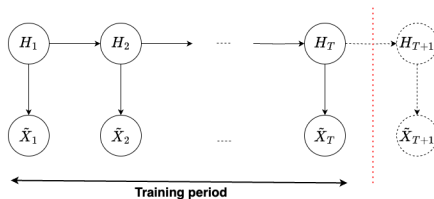
Predicting the next hidden state

Forecasting Market Turbulence Regimes

- ▶ After training the model with the EM algorithm, we want to predict the distribution over hidden states at the next time step $T + 1$.
- ▶ Objective: Compute

$$\forall h \in \llbracket 1, M \rrbracket, \quad p(H_{T+1} = h \mid \tilde{X}_1 = \tilde{x}_1, \dots, \tilde{X}_T = \tilde{x}_T)$$

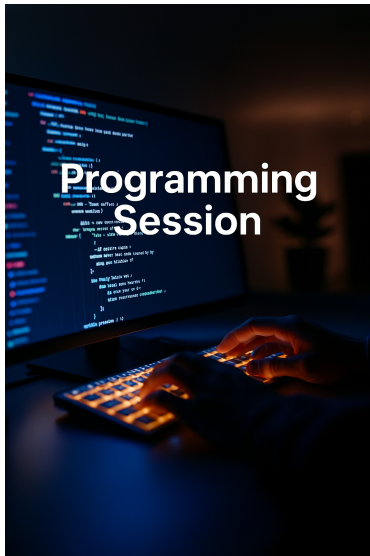
- ▶ This is useful for forecasting future regimes such as market turbulence.



- ▶ Using the filtering probabilities $\xi(T, \cdot)$, we derive for all $h \in \llbracket 1, M \rrbracket$:

$$\begin{aligned} p(H_{T+1} = h \mid \tilde{X}_{1:T}) &= \sum_{h'=1}^M p(H_{T+1} = h, H_T = h' \mid \tilde{X}_{1:T}) \\ &= \sum_{h'=1}^M \underbrace{p(H_{T+1} = h \mid H_T = h')}_{= Q_{h'h}} \cdot \underbrace{p(H_T = h' \mid \tilde{X}_{1:T})}_{= \xi(T, h')} \end{aligned}$$

- ▶ The filtering probabilities are computed using the Forward algorithm introduced earlier.



Programming Session 3: Section 3

- ▶ Section 3: Predicting the next Hidden State / Next Observation
- ▶ *Click here to access the programming session*

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- ▶ *Click here to access the GitHub Page*

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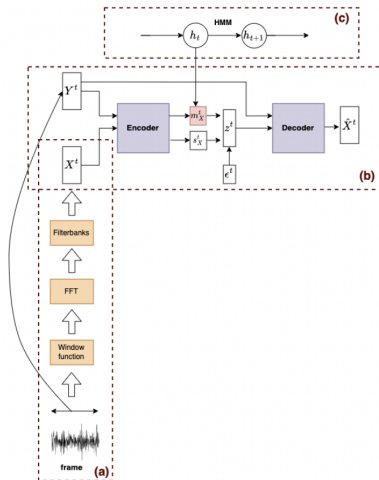
Forecasting Market Turbulence Regimes

Summary: The Low Turbulence Model processes return data to detect stable market regimes:

- ▶ Extract spectral information (Step a). [3]
- ▶ Learn low-dimensional structure (Step b). [1]
- ▶ Forecast regimes with HMMs (Step c). [4]

The model outputs regime probabilities used in asset allocation.

See [2] for full methodology and results.



Quiz Time!

Click here to take the quiz

Feedback Poll

Click here to participate in the poll

Thank you for your attention

- [1] Diederik P Kingma, Max Welling, et al. *Auto-encoding variational bayes*. 2013.
- [2] Hachem Madmoun. "Creating Investment Strategies Based on Machine Learning Algorithms". PhD thesis. École des Ponts ParisTech, 2022.
- [3] Stéphane Mallat. *A wavelet tour of signal processing*. Elsevier, 1999.
- [4] Lawrence R Rabiner. "A tutorial on hidden Markov models and selected applications in speech recognition". In: *Proceedings of the IEEE 77.2* (1989), pp. 257–286.